STAD29 / STA 1007 assignment 3

Due Tuesday Feb 5 at 11:59pm on Blackboard

Hand in the indicated questions. In preparation for the questions you hand in, it is worth your while to work through (or at least read through) the other questions as well.

Hand in your work on Quercus. If you did STAC32 last fall, it's the same procedure. A reminder is here: `https://www.utsc.utoronto.ca/~butler/c32/quercus1.nb.html`

You are reminded that work handed in with your name on it must be *entirely your own work*. It is as if you have signed your name under it. If it was done wholly or partly by someone else, *you have committed an academic offence*, and you can expect to be asked to explain yourself. The same applies if you allow someone else to copy your work. The grader will be watching out for assignments that look suspiciously similar to each other (or to my solutions). Besides which, if you do not do your own assignments, you *will* do badly on the exams, because the struggle to figure things out for yourself is an important part of the learning process.

Some packages that you will probably need too:

```
library(tidyverse)
library(nnet)
library(survival)
library(survminer)
```

Hand in questions 2 and 4 below.

1. Work through, or look at the problems in Chapter 17 of PASIAS that relate to nominal responses: 17.4, 17.5, 17.9, 17.11.

2. The US state of Maine has a lot of lakes. They are classified into three types, labelled 1, 2, and 3. (There is no particular significance to the use of these three numbers to label the lake types.)

   We have a number of measurements of mercury content in water samples (in parts per million) from lakes in Maine. For these water samples, the type of lake is known. Is it possible to build a model that would allow us to use a water sample from a lake of unknown type and predict what type of lake it is? Note that this is the opposite way around from ANOVA, where mercury content would be the response and lake type would be explanatory.

   The data are in `https://www.utsc.utoronto.ca/~butler/d29/mercury.txt` with the data values separated by a single space.

   (a) (2 marks) Read in the data and display at least some of your data frame. Do you have what you expected to have?

   (b) (2 marks) Why is this *not* the kind of model that you would fit with `polr`? Explain briefly.

   (c) (2 marks) Fit a suitable model for predicting lake type from mercury content. (We will investigate the model later.)

   (d) (3 marks) Test whether mercury content is significant in predicting lake type. What do you conclude?

   (e) (3 marks) For `mercury` values 0.1, 0.2, 0.5, 1, 1.5, predict lake type probabilities for each of them, and display your results side by side with the values they are predictions for.

   (f) (2 marks) Describe how predicted lake type changes as mercury content increases.

3. Study, or better, work through, Chapter 18 of PASIAS.

4. The `survival` package has several datasets in it. One is called `colon`, and comes from "one of the first successful trials of adjuvant chemotherapy" for colon cancer. You can look at the original data by loading the `survival` package and typing `colon` in a code chunk. You will see some missing values and some variables that are numeric codes (rather than actual numbers). Each subject also has two rows: a time to recurrence (if observed), and a time to death (if observed). I tidied up the data to include only the observations for recurrence (not death), to make all the categorical variables descriptive text rather than numeric codes, and to get rid of any subjects with any missing data. This data set can be found at `https://www.utsc.utoronto.ca/~butler/d29/colon.csv`. Use the data set at the link for the analysis in this question. Descriptions of the variables can be found in the help for the `colon` data set. To see that, type `?colon` in a code chunk, run it, and look in the bottom right pane.

   (a) (2 marks) What are the full names of the three treatments (as shown in the help for the data set)? The data set only contains abbreviations for these names. (I don't need to see output or screenshots here; if you get the right names, you must have gotten them from the right place.)

   (b) (2 marks) Read in and display some of the data. How many subjects do you have?

   (c) (3 marks) Run the following code, replacing my initial data frame `d` with the name of your data frame read in from the file:

   ```
   d %>% mutate_if(is.character, factor) %>% summary()
   ```

   Describe the output that it produces. There are two things to note: what it shows for a quantitative variable, and what it shows for a categorical variable.

   (d) (2 marks) Obtain a suitable response variable for a Cox proportional-hazards model, using the time until recurrence and the column that says whether colon cancer recurred or not for each subject. Display the first few values (say six) of your response variable.

   (e) (2 marks) The second value of your response variable (and also the 8th, 9th, and 10th, if you displayed that many) is displayed with a plus sign. What does that mean? (If you use a technical term, you have to explain what that technical term means.)

   (f) (3 marks) Fit a Cox proportional hazards model predicting the response variable you made in part (d) from all the other variables *except* for (i) those that were used in making the response variable and (ii) those that make no sense to include in a prediction model. You don't need to display your results. (Hint: in a model, `.` on the right side of the squiggle means "all the variables in the data frame except for the one used on the left of the squiggle, if that was in the data frame".)

   (g) (4 marks) Use `step` with `trace=0` to do a backward elimination of non-significant explanatory variables from the model of the previous part, one at a time. Save the result of `step` into a variable, and display the `drop1` output of your final model. Are all the remaining variables significant at $\alpha = 0.10$?

   Note: for the remainder of this question, use the output model from `step` as your "best" model. I don't need you to eliminate any more variables.

   (h) (3 marks) The major focus of the researchers was on whether the treatment affected the time to recurrence (of colon cancer). Is treatment in your output model from `step`?

   Use your results from part (c) to obtain "representative" values for all the other explanatory variables in that model: the median for quantitative variables, and the most frequently-observed category for categorical variables.

   (i) (2 marks) Make a data frame containing each of the three treatments combined with the representative values you obtained in the previous part. This is going to be an input to a prediction, so make sure the columns of your new data frame have the same names as the variables they are values of.

   (j) (2 marks) Obtain predicted survival curves for each of the three treatments and the representative values for the other variables that you obtained earlier. Bear in mind that we will be plotting the results shortly.

(k) (4 marks) Make a plot of the predicted survival curves for the three different treatments, and describe briefly how the three treatments compare. In your description, you should use the full names for the treatments, as far as they are known.

(l) (3 marks) Last part (finally!) Display the `summary` of your best model (the one that came out of `step`). How is (some part of) this output consistent with your graph? Explain briefly.

Notes

[1]This is rather in the same spirit as trying to determine who wrote a piece of text by looking at properties of the text like sentence length and use of certain words.

[2]Leaving out the `new` does predictions for the original data.

[3]There are also functions ending in `_at` that operate on columns that you select with a select-helper or any of the other ways you can run `select`.

[4]The hint here is that in the help for `colon` this column is tersely described as "censoring status".

[5]I could save it under the original name, and the one in my environment called `colon` would be the one that gets used, but it seems unnecessarily confusing to have two things called `colon` floating about.